

DSA Preparation

Array Important Question

<https://www.geeksforgeeks.org/problems/search-an-element-in-an-array-1587115621/1>

<https://leetcode.com/problems/sort-an-array/description/?envType=problem-list-v2&envId=array>

<https://leetcode.com/problems/find-peak-element/description/?envType=problem-list-v2&envId=array>

<https://leetcode.com/problems/rotate-array/description/?envType=problem-list-v2&envId=array>

<https://www.geeksforgeeks.org/problems/extract-the-integers4428/0>

<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/>

<https://leetcode.com/problems/majority-element/>

<https://leetcode.com/problems/product-of-array-except-self/description/>

<https://leetcode.com/problems/subarray-sum-equals-k/description/>

<https://leetcode.com/problems/contiguous-array/>

<https://leetcode.com/problems/non-overlapping-intervals/description/>

<https://leetcode.com/problems/insert-interval/description/>

String Important Question

<https://leetcode.com/problems/capitalize-the-title/description/>

<https://leetcode.com/problems/valid-palindrome/description/>

<https://leetcode.com/problems/reverse-words-in-a-string/description/>

<https://leetcode.com/problems/string-compression/description/>

<https://leetcode.com/problems/reverse-only-letters/description/>

<https://leetcode.com/problems/longest-common-prefix/description/>

<https://leetcode.com/problems/longest-palindrome/description/>

<https://leetcode.com/problems/string-to-integer-atoi/description/>

HashMap Important Question

<https://leetcode.com/problems/intersection-of-two-arrays/description/>

<https://leetcode.com/problems/single-number/description/>

https://www.geeksforgeeks.org/problems/find-distinct-elements--130928/1?itm_source=geeksforgeeks&itm_medium=article&itm_campaign=practice_card

<https://leetcode.com/problems/ransom-note/description/>

<https://leetcode.com/problems/insert-delete-getrandom-o1/description/>

<https://leetcode.com/problems/longest-consecutive-sequence/>

<https://leetcode.com/problems/rotate-image/description/>

<https://leetcode.com/problems/spiral-matrix/description/>

<https://leetcode.com/problems/valid-sudoku/description/>

<https://leetcode.com/problems/set-matrix-zeroes/description/>

<https://leetcode.com/problems/magic-squares-in-grid/>

Matrix Important Question

<https://leetcode.com/problems/minimum-absolute-difference/description/?envType=problem-list-v2&envId=sorting>

<https://leetcode.com/problems/wiggle-sort-ii/description/?envType=problem-list-v2&envId=sorting>

<https://leetcode.com/problems/sort-colors/description/>

<https://leetcode.com/problems/first-missing-positive/description/>

<https://leetcode.com/problems/missing-number/>

<https://leetcode.com/problems/find-all-duplicates-in-an-array/>

<https://leetcode.com/problems/move-zeroes/>

<https://leetcode.com/problems/first-missing-positive/description/>

Sorting Important Question

<https://leetcode.com/problems/merge-sorted-array/>

<https://leetcode.com/problems/minimum-absolute-difference/description/?envType=problem-list-v2&envId=sorting>

<https://leetcode.com/problems/wiggle-sort-ii/description/?envType=problem-list-v2&envId=sorting>

<https://leetcode.com/problems/sort-colors/description/>

<https://leetcode.com/problems/first-missing-positive/description/>

<https://leetcode.com/problems/first-missing-positive/description/>

<https://leetcode.com/problems/move-zeroes/description/>

<https://leetcode.com/problems/find-all-duplicates-in-an-array/description/>

<https://leetcode.com/problems/missing-number/description/>

Binary Search Tree

<http://leetcode.com/problems/search-in-rotated-sorted-array/description/>

<https://leetcode.com/problems/search-a-2d-matrix/description/>

<https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/description/>

<https://leetcode.com/problems/first-bad-version/description/>

<https://leetcode.com/problems/sqrtx/description/>

<https://leetcode.com/problems/find-peak-element/description/>

<https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/description/>

<https://leetcode.com/problems/find-the-duplicate-number/description/>

<https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/description/>

<https://leetcode.com/problems/single-element-in-a-sorted-array/description/>

<https://leetcode.com/problems/find-smallest-letter-greater-than-target/description/>

Two Pointers

<http://leetcode.com/problems/two-sum/description/>

<https://leetcode.com/problems/3sum/description/>

<https://leetcode.com/problems/trapping-rain-water/description/>

<https://leetcode.com/problems/container-with-most-water/description/>

<https://leetcode.com/problems/valid-anagram/description/>
<https://leetcode.com/problems/squares-of-a-sorted-array/description/>
<https://leetcode.com/problems/longest-palindromic-substring/description/>

Stack

<https://leetcode.com/problems/evaluate-reverse-polish-notation/description/>
<https://leetcode.com/problems/basic-calculator/description/>
<https://leetcode.com/problems/longest-valid-parentheses/description/>
<https://leetcode.com/problems/min-stack/description/>
<https://leetcode.com/problems/valid-parentheses/description/>
<http://leetcode.com/problems/decode-string/description/>

Monotonic Stack

<https://www.geeksforgeeks.org/problems/next-larger-element-1587115620/1>
<https://leetcode.com/problems/daily-temperatures/description/>
<https://leetcode.com/problems/largest-rectangle-in-histogram/description/>
<https://leetcode.com/problems/asteroid-collision/description/>
<https://leetcode.com/problems/implement-queue-using-stacks/description/>
<https://leetcode.com/problems/backspace-string-compare/description/>

Queue

<https://leetcode.com/problems/design-circular-queue/description/>
<https://leetcode.com/problems/sliding-window-maximum/description/>
<https://leetcode.com/problems/implement-stack-using-queues/description/>

Linked List

<https://leetcode.com/problems/copy-list-with-random-pointer/description/>
<https://leetcode.com/problems/merge-two-sorted-lists/description/>
<https://leetcode.com/problems/reverse-linked-list/description/>
<https://leetcode.com/problems/middle-of-the-linked-list/description/>
<https://leetcode.com/problems/remove-nth-node-from-end-of-list/description/>
<https://leetcode.com/problems/find-the-duplicate-number/description/>
<https://leetcode.com/problems/linked-list-cycle/description/>
<https://leetcode.com/problems/sort-list/description/>
<https://leetcode.com/problems/partition-list/description/>
<https://leetcode.com/problems/reorder-list/description/>
<https://leetcode.com/problems/palindrome-linked-list/description/>
<https://leetcode.com/problems/swap-nodes-in-pairs/description/>
<https://leetcode.com/problems/odd-even-linked-list/description/>
<https://www.geeksforgeeks.org/problems/add-1-to-a-number-represented-as-linked-list/1>
<https://leetcode.com/problems/intersection-of-two-linked-lists/description/>
<https://leetcode.com/problems/delete-the-middle-node-of-a-linked-list/description/>
<https://leetcode.com/problems/linked-list-cycle-ii/description/>

<https://leetcode.com/problems/flatten-a-multilevel-doubly-linked-list/description/>
<https://leetcode.com/problems/copy-list-with-random-pointer/description/>
https://www.geeksforgeeks.org/problems/delete-all-occurrences-of-a-given-key-in-a-doubly-linked-list/1?utm_source=chatgpt.com

Recursion

<https://leetcode.com/problems/generate-parentheses/description/>
<https://leetcode.com/problems/letter-combinations-of-a-phone-number/description/>
<http://geeksforgeeks.org/problems/tower-of-hanoi-1587115621/1>
<https://leetcode.com/problems/next-permutation/description/>

Backtracking

<https://leetcode.com/problems/permutations/description/>
<https://leetcode.com/problems/subsets/description/>
<https://leetcode.com/problems/combination-sum/description/>
<https://leetcode.com/problems/sudoku-solver/description/>
<https://leetcode.com/problems/n-queens/description/>

Heap

<https://leetcode.com/problems/k-closest-points-to-origin/description/>
<https://leetcode.com/problems/kth-largest-element-in-an-array/description/>
<https://leetcode.com/problems/top-k-frequent-words/description/>
<https://leetcode.com/problems/find-k-closest-elements/description/>
<https://leetcode.com/problems/task-scheduler/description/>

Sliding Window

<https://leetcode.com/problems/maximum-subarray/description/>
<https://leetcode.com/problems/longest-substring-without-repeating-characters/description/>
<https://leetcode.com/problems/longest-repeating-character-replacement/description/>
<https://leetcode.com/problems/minimum-window-substring/description/>
<https://leetcode.com/problems/find-all-anagrams-in-a-string/description/>
<https://leetcode.com/problems/sliding-window-maximum/description/>

Tree-1

<https://leetcode.com/problems/invert-binary-tree/description/>
<https://leetcode.com/problems/same-tree/description/>
<https://leetcode.com/problems/symmetric-tree/description/>
<https://leetcode.com/problems/subtree-of-another-tree/description/>
<https://leetcode.com/problems/maximum-depth-of-binary-tree/description/>
<https://leetcode.com/problems/balanced-binary-tree/description/>
<https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/description/>

Tree-2

<https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-tree/description/>
<https://leetcode.com/problems/kth-smallest-element-in-a-bst/description/>
<https://leetcode.com/problems/validate-binary-search-tree/description/>
<https://leetcode.com/problems/serialize-and-deserialize-binary-tree/description/>
<https://leetcode.com/problems/maximum-width-of-binary-tree/description/>
<https://leetcode.com/problems/binary-tree-maximum-path-sum/description/>

Graph-1

<https://leetcode.com/problems/flood-fill/description/>
<https://leetcode.com/problems/flood-fill/description/>
<https://leetcode.com/problems/01-matrix/description/>
<https://leetcode.com/problems/clone-graph/description/>
<https://leetcode.com/problems/number-of-islands/description/>
<https://leetcode.com/problems/rotting-oranges/description/>
<https://leetcode.com/problems/word-search/description/>
<https://leetcode.com/problems/word-ladder/description/>

Graph-2

<https://leetcode.com/problems/pacific-atlantic-water-flow/description/>
<https://www.geeksforgeeks.org/problems/topological-sort/0>
<https://leetcode.com/problems/course-schedule/description/>
<https://leetcode.com/problems/course-schedule-ii/description/>
<https://www.geeksforgeeks.org/problems/prerequisite-tasks/1>
<https://leetcode.com/problems/number-of-provinces/description/>
<https://www.geeksforgeeks.org/problems/connected-components-in-an-undirected-graph/1>
<https://leetcode.com/problems/longest-increasing-path-in-a-matrix/description/>
<https://leetcode.com/problems/cheapest-flights-within-k-stops/description/>
<https://www.geeksforgeeks.org/problems/alien-dictionary/1>
<https://leetcode.com/problems/accounts-merge/description/>

Greedy

<https://leetcode.com/problems/gas-station/description/>
<https://leetcode.com/problems/largest-number/description/>
<https://leetcode.com/problems/merge-intervals/description/>
<https://www.geeksforgeeks.org/problems/attend-all-meetings/1>

Dynamic Programming

<https://leetcode.com/problems/fibonacci-number/description/>
<https://leetcode.com/problems/climbing-stairs/description/>
<https://leetcode.com/problems/decode-ways/description/>
<https://leetcode.com/problems/maximum-product-subarray/description/>
<https://leetcode.com/problems/house-robber/description/>
<https://leetcode.com/problems/jump-game/description/>

<https://leetcode.com/problems/count-square-submatrices-with-all-ones/description/>
<https://leetcode.com/problems/maximal-square/description/>
<https://leetcode.com/problems/coin-change/description/>
<https://www.geeksforgeeks.org/problems/subset-sum-problem-1611555638/1>
<https://leetcode.com/problems/partition-equal-subset-sum/description/>
<https://leetcode.com/problems/longest-increasing-subsequence/description/>

Tries

<https://leetcode.com/problems/implement-trie-prefix-tree/description/>
<https://leetcode.com/problems/word-break/description/>
<https://leetcode.com/problems/design-add-and-search-words-data-structure/description/>

Bits

<https://leetcode.com/problems/single-number/description/>
<https://leetcode.com/problems/find-the-duplicate-number/description/>
<https://leetcode.com/problems/counting-bits/description/>
<https://leetcode.com/problems/number-of-1-bits/description/>
<https://leetcode.com/problems/missing-number/description/>
<https://leetcode.com/problems/reverse-bits/description/>
<https://leetcode.com/problems/add-binary/description/>

Maths

<https://leetcode.com/problems/roman-to-integer/description/>
<https://leetcode.com/problems/powx-n/description/>
<https://leetcode.com/problems/reverse-integer/description/>
<https://leetcode.com/problems/palindrome-number/description/>
<https://leetcode.com/problems/count-primes/description/>

Design Problems

<https://leetcode.com/problems/min-stack/description/>
<https://leetcode.com/problems/find-median-from-data-stream/description/>
<https://leetcode.com/problems/design-add-and-search-words-data-structure/description/>
<https://leetcode.com/problems/design-circular-queue/description/>
<https://leetcode.com/problems/find-median-from-data-stream/description/>
<https://leetcode.com/problems/time-based-key-value-store/description/>
<https://leetcode.com/problems/lru-cache/description/>
<https://leetcode.com/problems/insert-delete-getrandom-o1/description/>
<https://leetcode.com/problems/implement-trie-prefix-tree/description/>

Algorithm



Algorithms



Quick Sort Algorithm

0:28:54



Prims & Kruskal Algorithm

0:30:31



Dijkstra Algorithm

0:30:19



Bellman Ford Algorithm

0:32:54



Floyd Warshal Algorithm

0:20:21